JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Functions

JEE Main Mathematics · Previous Year Questions with Answer Key

175	128	2002 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Numerical	35 questions
MCQ (Single Correct Answer)	140 questions
Answer key	at the end

1 **Solve**
Print it, or work through it on screen.

2 **Check**
The answer key is on the last pages.

Numerical 35 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Let f be a polynomial function such that $\log_2(f(x)) = \left(\log_2\left(2+\frac{2}{3}+\frac{2}{9}+\ldots\ldots\infty\right)\right)\cdot\log_3\left(1+\frac{f(x)}{f(1/x)}\right), x>0$ and $f(6)=37$. Then $\sum_{n=1}^{10} f(n)$ is equal to ____ .

Q2. JEE Main 2026 (Online) 5th April Morning Shift

Let $A=\{1,2,3,4,5,6\}$. The number of one-one functions $f: A \to A$ such that $f(1)\geq 3, f(3)\leq 4$ and $f(2)+f(3)=5$, is ____ .

Q3. JEE Main 2026 (Online) 2nd April Morning Shift

If the domain of the function

$f(x)=\sqrt{\log_{(0.6)}\left(\left|\frac{2x-5}{x^2-4}\right|\right)}$ is $(-\infty,a]\cup\{b\}\cup[c,d)\cup(e,\infty)$, then the value of $a+b+c+d+e$ is _____.

Q4. JEE Main 2025 (Online) 8th April Evening Shift

Let the domain of the function $f(x)=\cos^{-1}\left(\frac{4x+5}{3x-7}\right)$ be $[\alpha,\beta]$ and the domain of $g(x)=\log_2(2-6\log_{27}(2x+5))$ be (γ,δ).

Then $|7(\alpha+\beta)+4(\gamma+\delta)|$ is equal to _____.

Q5. JEE Main 2024 (Online) 9th April Evening Shift

Let $A=\{(x,y): 2x+3y=23, x,y\in\mathbb{N}\}$ and $B=\{x:(x,y)\in A\}$. Then the number of one-one functions from A to B is equal to _____.

Q6. JEE Main 2024 (Online) 9th April Morning Shift

If a function f satisfies $f(m+n)=f(m)+f(n)$ for all $m,n\in\mathbf{N}$ and $f(1)=1$, then the largest natural number λ such that $\sum_{k=1}^{2022} f(\lambda+k)\leq(2022)^2$ is equal to _____.

Q7. JEE Main 2024 (Online) 8th April Morning Shift

If the range of $f(\theta)=\frac{\sin^4\theta+3\cos^2\theta}{\sin^4\theta+\cos^2\theta}, \theta\in\mathbb{R}$ is $[\alpha,\beta]$, then the sum of the infinite G.P., whose first term is 64 and the common ratio is $\frac{\alpha}{\beta}$, is equal to _____.

Q8. JEE Main 2024 (Online) 5th April Morning Shift

If $S=\{a \in \mathbf{R}:|2a-1|=3[a]+2\{a\}\}$, where $[t]$ denotes the greatest integer less than or equal to t and $\{t\}$ represents the fractional part of t, then $72 \sum\limits_{a \in S} a$ is equal to _____.

Q9. JEE Main 2024 (Online) 4th April Evening Shift

Consider the function $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x)=\frac{2x}{\sqrt{1+9x^2}}$. If the composition of $\underbrace{(f \circ f \circ f \circ \cdots \circ f)}_{10 \text{ times}}(x)=\frac{2^{10}x}{\sqrt{1+9\alpha x^2}}$, then the value of $\sqrt{3\alpha+1}$ is equal to _____.

Q10. JEE Main 2024 (Online) 30th January Morning Shift

Let $\mathrm{A}=\{1,2,3,\ldots,7\}$ and let $\mathrm{P}(\mathrm{A})$ denote the power set of A. If the number of functions $f: \mathrm{A} \to \mathrm{P}(\mathrm{A})$ such that $\mathrm{a} \in f(\mathrm{a}), \forall \mathrm{a} \in \mathrm{A}$ is $\mathrm{m}^{\mathrm{n}}, \mathrm{m}$ and $\mathrm{n} \in \mathrm{N}$ and m is least, then $\mathrm{m}+\mathrm{n}$ is equal to _____.

Q11. JEE Main 2023 (Online) 11th April Evening Shift

Let $\mathrm{A}=\{1,2,3,4,5\}$ and $\mathrm{B}=\{1,2,3,4,5,6\}$. Then the number of functions $f: \mathrm{A} \to \mathrm{B}$ satisfying $f(1)+f(2)=f(4)-1$ is equal to _____.

Q12. JEE Main 2023 (Online) 8th April Evening Shift

Let $\mathrm{R}=\{\mathrm{a},\mathrm{b},\mathrm{c},\mathrm{d},\mathrm{e}\}$ and $\mathrm{S}=\{1,2,3,4\}$. Total number of onto functions $f: \mathrm{R} \to \mathrm{S}$ such that $f(\mathrm{a}) \neq 1$, is equal to _____.

Q13. JEE Main 2023 (Online) 8th April Evening Shift

If domain of the function $\log_e\left(\frac{6x^2+5x+1}{2x-1}\right)+\cos^{-1}\left(\frac{2x^2-3x+4}{3x-5}\right)$ is $(\alpha,\beta) \cup (\gamma,\delta]$, then $18\left(\alpha^2+\beta^2+\gamma^2+\delta^2\right)$ is equal to _____.

Q14. JEE Main 2023 (Online) 30th January Evening Shift

Let $A=\{1,2,3,5,8,9\}$. Then the number of possible functions $f: A \to A$ such that $f(m \cdot n)=f(m) \cdot f(n)$ for every $m,n \in A$ with $m \cdot n \in A$ is equal to _____.

Q15. JEE Main 2023 (Online) 30th January Morning Shift

Let $S=\{1,2,3,4,5,6\}$. Then the number of one-one functions $f: \mathrm{S} \to \mathrm{P}(\mathrm{S})$, where $\mathrm{P}(\mathrm{S})$ denote the power set of S, such that $f(n) \subset f(\mathrm{m})$ where $n<m$ is _____.

Q16. JEE Main 2023 (Online) 30th January Morning Shift

Let $f^1(x)=\frac{3x+2}{2x+3}, x \in \mathbf{R}-\left\{\frac{-3}{2}\right\}$ For $\mathrm{n} \geq 2$, define $f^{\mathrm{n}}(x)=f^1 o f^{\mathrm{n}-1}(x)$. If $f^5(x)=\frac{\mathrm{a}x+\mathrm{b}}{\mathrm{b}x+\mathrm{a}}, \gcd(\mathrm{a}, \mathrm{b})=1$, then $\mathrm{a}+\mathrm{b}$ is equal to _____.

Q17. JEE Main 2023 (Online) 29th January Morning Shift

Suppose f is a function satisfying $f(x+y)=f(x)+f(y)$ for all $x, y \in N$ and $f(1)=\frac{1}{5}$. If $\sum_{n=1}^{m} \frac{f(n)}{n(n+1)(n+2)}=\frac{1}{12}$, then m is equal to _____.

Q18. JEE Main 2023 (Online) 25th January Morning Shift

For some a, b, $\mathrm{c} \in \mathbb{N}$, let $f(x)=ax-3$ and $\mathrm{g}(\mathrm{x})=\mathrm{x}^{\mathrm{b}}+\mathrm{c}, \mathrm{x} \in \mathbb{R}$. If $(fog)^{-1}(x)=\left(\frac{x-7}{2}\right)^{1/3}$, then $(fog)(ac)+(gof)(b)$ is equal to _____.

Q19. JEE Main 2022 (Online) 28th July Morning Shift

For $\mathrm{p}, \mathrm{q} \in \mathbf{R}$, consider the real valued function $f(x)=(x-\mathrm{p})^2-\mathrm{q}, x \in \mathbf{R}$ and $\mathrm{q}>0$. Let $\mathrm{a}_1, \mathrm{a}_2, \mathrm{a}_3$ and a_4 be in an arithmetic progression with mean p and positive common difference. If $\left|f\left(\mathrm{a}_i\right)\right|=500$ for all $i=1,2,3,4$, then the absolute difference between the roots of $f(x)=0$ is _____.

Q20. JEE Main 2022 (Online) 27th July Evening Shift

The number of functions f, from the set $\mathrm{A}=\left\{x \in \mathbf{N}: x^2-10x+9 \leq 0\right\}$ to the set $\mathrm{B}=\left\{\mathrm{n}^2: \mathrm{n} \in \mathbf{N}\right\}$ such that $f(x) \leq (x-3)^2+1$, for every $x \in \mathrm{A}$, is _____.

Q21. JEE Main 2022 (Online) 27th July Morning Shift

Let $f(x)=2x^2-x-1$ and $\mathrm{S}=\{n \in \mathbb{Z}:|f(n)| \leq 800\}$. Then, the value of $\sum_{n \in S} f(n)$ is equal to _____.

Q22. JEE Main 2022 (Online) 25th July Evening Shift

Let $f(x)$ be a quadratic polynomial with leading coefficient 1 such that $f(0)=p, p \neq 0$, and $f(1)=\frac{1}{3}$. If the equations $f(x)=0$ and $f \circ f \circ f \circ f(x)=0$ have a common real root, then $f(-3)$ is equal to _____.

Q23. JEE Main 2022 (Online) 29th June Evening Shift

Let f(x) and g(x) be two real polynomials of degree 2 and 1 respectively. If $f(g(x))=8x^2-2x$ and $g(f(x))=4x^2+6x+1$, then the value of $f(2)+g(2)$ is _____.

Q24. JEE Main 2022 (Online) 29th June Morning Shift

Let c, k ∈ R. If $f(x) = (c+1)x^2 + (1-c^2)x + 2k$ and $f(x+y) = f(x) + f(y) - xy$, for all x, y ∈ R, then the value of $|2(f(1) + f(2) + f(3) + \ldots\ldots + f(20))|$ is equal to _____.

Q25. JEE Main 2022 (Online) 28th June Evening Shift

Let S = {1, 2, 3, 4}. Then the number of elements in the set { f : S × S → S : f is onto and f (a, b) = f (b, a) ≥ a ∀ (a, b) ∈ S × S } is _____.

Q26. JEE Main 2022 (Online) 27th June Evening Shift

Let S = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}. Define f : S → S as

$f(n) = \begin{cases} 2n & , \quad if\, n = 1,2,3,4,5 \\ 2n-11 & , \quad if\, n = 6,7,8,9,10 \end{cases}$.

Let g : S → S be a function such that $fog(n) = \begin{cases} n+1 & , \quad if\, n\ is\ odd \\ n-1 & , \quad if\, n\ is\ even \end{cases}$.

Then $g(10)(g(1) + g(2) + g(3) + g(4) + g(5))$ is equal to _____.

Q27. JEE Main 2022 (Online) 27th June Morning Shift

Let f : R → R be a function defined by $f(x) = \frac{2e^{2x}}{e^{2x}+e}$. Then

$f\left(\frac{1}{100}\right) + f\left(\frac{2}{100}\right) + f\left(\frac{3}{100}\right) + \ldots\ldots + f\left(\frac{99}{100}\right)$ is equal to _____.

Q28. JEE Main 2022 (Online) 25th June Morning Shift

Let $f : R \to R$ be a function defined by

$f(x) = \left(2\left(1 - \frac{x^{25}}{2}\right)(2 + x^{25})\right)^{\frac{1}{50}}$. If the function $g(x) = f(f(f(x))) + f(f(x))$, then the greatest integer less than or equal to g(1) is _____.

Q29. JEE Main 2022 (Online) 24th June Morning Shift

The number of one-one functions f : {a, b, c, d} → {0, 1, 2, ..., 10} such

that 2f(a) − f(b) + 3f(c) + f(d) = 0 is _____.

Q30. JEE Main 2021 (Online) 27th July Morning Shift

Let S = {1, 2, 3, 4, 5, 6, 7}. Then the number of possible functions f : S → S
such that f(m . n) = f(m) . f(n) for every m, n ∈ S and m . n ∈ S is equal to _____.

Q31. JEE Main 2021 (Online) 22th July Evening Shift

Let A = {0, 1, 2, 3, 4, 5, 6, 7}. Then the number of bijective functions f : A $\rightarrow$ A such that f(1) + f(2) = 3 $-$ f(3) is equal to

Q32. JEE Main 2021 (Online) 18th March Evening Shift

If f(x) and g(x) are two polynomials such that the polynomial P(x) = $f(x^3)$ + x $g(x^3)$ is divisible by x^2 + x + 1, then P(1) is equal to _____.

Q33. JEE Main 2021 (Online) 24th February Evening Shift

If a + α = 1, b + β = 2 and $af(x) + \alpha f\left(\frac{1}{x}\right) = bx + \frac{\beta}{x}, x \neq 0$, then the value of the expression $\frac{f(x)+f\left(\frac{1}{x}\right)}{x+\frac{1}{x}}$ is _____.

Q34. JEE Main 2020 (Online) 6th September Evening Slot

Suppose that a function f : R $\rightarrow$ R satisfies
f(x + y) = f(x)f(y) for all x, y $\in$ R and f(1) = 3.
If $\sum_{i=1}^{n} f(i) = 363$ then n is equal to _____ .

Q35. JEE Main 2020 (Online) 5th September Evening Slot

Let A = {a, b, c} and B = {1, 2, 3, 4}. Then the number of elements in the set
C = {f : A $\rightarrow$ B | 2 $\in$ f(A) and f is not one-one} is _____.

MCQ (Single Correct Answer) 140 questions

Q36. JEE Main 2026 (Online) 6th April Evening Shift

Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be defined as $f(x) = \frac{2x^2-3x+2}{3x^2+x+3}$. Then f is :

(A) both one-one and onto
(B) one-one but not onto
(C) onto but not one-one
(D) neither one-one nor onto

Q37. JEE Main 2026 (Online) 6th April Morning Shift

Let [•] denote the greatest integer function. If the domain of the function $f(x) = \sin^{-1}\left(\frac{x+[x]}{3}\right)$ is $[\alpha, \beta)$, then $\alpha^2 + \beta^2$ is equal to:

(A) 2
(B) 5
(C) 10
(D) 13

Q38. JEE Main 2026 (Online) 4th April Evening Shift

For the function $f : [1, \infty) \to [1, \infty)$ defined by $f(x) = (x-1)^4 + 1$, among the two statements:

(I) The set $S = \{x \in [1, \infty) : f(x) = f^{-1}(x)\}$ contains exactly two elements, and

(II) The set $S = \{x \in [1, \infty) : f(x) = f^{-1}(x+1)\}$ is an empty set,

(A) only (I) is TRUE

(B) only (II) is TRUE

(C) both (I) and (II) are TRUE

(D) neither (I) nor (II) is TRUE

Q39. JEE Main 2026 (Online) 4th April Evening Shift

Let for some $\alpha \in \mathbb{R}$, $f : \mathbb{R} \to \mathbb{R}$ be a function satisfying $f(x+y) = f(x) + 2y^2 + y + \alpha xy$ for all $x, y \in \mathbb{R}$. If $f(0) = -1$ and $f(1) = 2$, then the value of $\sum_{n=1}^{5} (\alpha + f(n))$ is :

(A) 110

(B) 140

(C) 150

(D) 170

Q40. JEE Main 2026 (Online) 4th April Morning Shift

Let [•] denote the greatest integer function. If the domain of the function

$f(x) = \cos^{-1}\left(\frac{4x+2[x]}{3}\right)$ is $[\alpha, \beta]$, then $12(\alpha + \beta)$ is equal to :

(A) 6

(B) 8

(C) 9

(D) 4

Q41. JEE Main 2026 (Online) 4th April Morning Shift

The number of functions $f : \{1, 2, 3, 4\} \to \{a, b, c\}$, which are not onto, is :

(A) 48

(B) 45

(C) 51

(D) 35

Q42. JEE Main 2026 (Online) 28th January Evening Shift

Given below are two statements :

Statement I : The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \frac{x}{1+|x|}$ is one-one.

Statement II : The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \frac{x^2+4x-30}{x^2-8x+18}$ is many-one.

In the light of the above statements, choose the **correct answer** from the options given below :

(A) **Statement I** is true but **Statement II** is false

(B) Both **Statement I** and **Statement II** are false

(C) Both **Statement I** and **Statement II** are true

(D) **Statement I** is false but **Statement II** is true

Q43. JEE Main 2026 (Online) 28th January Evening Shift

The sum of all the elements in the range of $f(x) = \text{Sgn}(\sin x) + \text{Sgn}(\cos x) + \text{Sgn}(\tan x) + \text{Sgn}(\cot x)$, $x \neq \frac{n\pi}{2}, n \in \mathbb{Z}$, where

$\text{Sgn}(t) = \begin{cases} 1, & \text{if } t > 0 \\ -1, & \text{if } t < 0 \end{cases}$

is :

(A) 4 (B) 0 (C) 2 (D) -2

Q44. JEE Main 2026 (Online) 28th January Morning Shift

If $g(x) = 3x^2 + 2x - 3$, $f(0) = -3$ and $4g(f(x)) = 3x^2 - 32x + 72$, then $f(g(2))$ is equal to:

(A) $\frac{7}{2}$ (B) $-\frac{25}{6}$ (C) $\frac{25}{6}$ (D) $-\frac{7}{2}$

Q45. JEE Main 2026 (Online) 24th January Evening Shift

Let f be a function such that $3f(x) + 2f\left(\frac{m}{19x}\right) = 5x, x \neq 0$, where $m = \sum_{i=1}^{9}(i)^2$. Then $f(5) - f(2)$ is equal to

(A) 36 (B) 9 (C) -9 (D) 18

Q46. JEE Main 2026 (Online) 24th January Morning Shift

If the domain of the function

$f(x) = \log_{(10x^2-17x+7)}\left(18x^2 - 11x + 1\right)$

is $(-\infty, a) \cup (b, c) \cup (d, \infty) - \{e\}$, then

$90(a + b + c + d + e)$ equals:

(A) 170 (B) 316 (C) 177 (D) 307

Q47. JEE Main 2026 (Online) 23rd January Evening Shift

Consider two sets $\text{A} = \{x \in \text{Z} : |(|x - 3| - 3)| \leq 1\}$ and

$\text{B} = \left\{x \in \mathbb{R} - \{1, 2\} : \frac{(x-2)(x-4)}{x-1}\log_e(|x - 2|) = 0\right\}$. Then the number of

onto functions $f : \text{A} \to \text{B}$ is equal to :

(A) 32 (B) 81 (C) 79 (D) 62

Q48. JEE Main 2026 (Online) 22nd January Evening Shift

Let $f(x)=[x]^2-[x+3]-3, x \in \mathbf{R}$, where [.] is the greatest integer function. Then

(A) $f(x)=0$ for finitely many values of x

(B) $f(x)<0$ only for $x \in [-1,3)$

(C) $\int_0^2 f(x)\mathrm{d}x=-6$

(D) $f(x)>0$ only for $x \in [4,\infty)$

Q49. JEE Main 2026 (Online) 22nd January Evening Shift

Let the domain of the function $f(x)=\log_3 \log_5 \left(7-\log_2\left(x^2-10x+85\right)\right)+\sin^{-1}\left(\left|\frac{3x-7}{17-x}\right|\right)$ be $(\alpha, \beta]$. Then $\alpha+\beta$ is equal to :

(A) 12 (B) 8 (C) 10 (D) 9

Q50. JEE Main 2026 (Online) 22nd January Evening Shift

Let f and g be functions satisfying $f(x+y)=f(x)f(y), f(1)=7$ and $g(x+y)=g(xy), g(1)=1$, for all $x, y \in \mathbf{N}$. If $\sum_{x=1}^{n}\left(\frac{f(x)}{g(x)}\right)=19607$, then n is equal to :

(A) 6 (B) 7 (C) 4 (D) 5

Q51. JEE Main 2026 (Online) 22nd January Morning Shift

If the domain of the function $f(x)=\sin^{-1}\left(\frac{5-x}{3+2x}\right)+\frac{1}{\log_e(10-x)}$ is $(-\infty, \alpha] \cup [\beta, \gamma)-\{\delta\}$, then $6(\alpha+\beta+\gamma+\delta)$ is equal to

(A) 66 (B) 68 (C) 70 (D) 67

Q52. JEE Main 2025 (Online) 7th April Evening Shift

If the range of the function $f(x)=\frac{5-x}{x^2-3x+2}$, $x \neq 1,2$, is $(-\infty, \alpha] \cup [\beta, \infty)$, then $\alpha^2+\beta^2$ is equal to :

(A) 188 (B) 192 (C) 190 (D) 194

Q53. JEE Main 2025 (Online) 4th April Evening Shift

Let the domains of the functions $f(x)=\log_4 \log_3 \log_7\left(8-\log_2\left(x^2+4x+5\right)\right)$ and $g(x)=\sin^{-1}\left(\frac{7x+10}{x-2}\right)$ be (α, β) and $[\gamma, \delta]$, respectively. Then $\alpha^2+\beta^2+\gamma^2+\delta^2$ is equal to :

(A) 15 (B) 13 (C) 16 (D) 14

Q54. JEE Main 2025 (Online) 4th April Morning Shift

Let $f, g:(1,\infty) \to \mathbb{R}$ be defined as $f(x)=\frac{2x+3}{5x+2}$ and $g(x)=\frac{2-3x}{1-x}$. If the range of the function $f \circ g:[2,4] \to \mathbb{R}$ is $[\alpha, \beta]$, then $\frac{1}{\beta-\alpha}$ is equal to

(A) 56 (B) 2 (C) 29 (D) 68

Q55. JEE Main 2025 (Online) 3rd April Evening Shift

Let f be a function such that $f(x)+3f\left(\frac{24}{x}\right)=4x, x\neq 0$. Then $f(3)+f(8)$ is equal to

(A) 13 (B) 11 (C) 10 (D) 12

Q56. JEE Main 2025 (Online) 3rd April Evening Shift

If the domain of the function $f(x)=\log_7\left(1-\log_4\left(x^2-9x+18\right)\right)$ is $(\alpha,\beta)\cup(\gamma,\delta)$, then $\alpha+\beta+\gamma+\delta$ is equal to

(A) 17 (B) 15 (C) 16 (D) 18

Q57. JEE Main 2025 (Online) 3rd April Morning Shift

If the domain of the function $f(x)=\log_e\left(\frac{2x-3}{5+4x}\right)+\sin^{-1}\left(\frac{4+3x}{2-x}\right)$ is $[\alpha,\beta)$, then $\alpha^2+4\beta$ is equal to

(A) 4 (B) 3 (C) 7 (D) 5

Q58. JEE Main 2025 (Online) 2nd April Evening Shift

If the domain of the function $f(x)=\frac{1}{\sqrt{10+3x-x^2}}+\frac{1}{\sqrt{x+|x|}}$ is (a,b), then $(1+a)^2+b^2$ is equal to :

(A) 29 (B) 30 (C) 25 (D) 26

Q59. JEE Main 2025 (Online) 29th January Evening Shift

If the domain of the function $\log_5(18x-x^2-77)$ is (α,β) and the domain of the function $\log_{(x-1)}\left(\frac{2x^2+3x-2}{x^2-3x-4}\right)$ is (γ,δ), then $\alpha^2+\beta^2+\gamma^2$ is equal to:

(A) 186 (B) 179 (C) 195 (D) 174

Q60. JEE Main 2025 (Online) 28th January Evening Shift

Let $f:[0,3]\to$ A be defined by $f(x)=2x^3-15x^2+36x+7$ and $g:[0,\infty)\to B$ be defined by $g(x)=\frac{x^{2025}}{x^{2025}+1}$, If both the functions are onto and $S=\{x\in Z; x\in A \text{ or } x\in B\}$, then $n(S)$ is equal to :

(A) 29 (B) 31 (C) 30 (D) 36

Q61. JEE Main 2025 (Online) 28th January Morning Shift

If $f(x)=\frac{2^x}{2^x+\sqrt{2}}, \mathrm{x}\in\mathbb{R}$, then $\sum_{\mathrm{k}=1}^{81} f\left(\frac{\mathrm{k}}{82}\right)$ is equal to

(A) 82 (B) $81\sqrt{2}$ (C) 41 (D) $\frac{81}{2}$

Q62. JEE Main 2025 (Online) 28th January Morning Shift

Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = (2 + 3a)x^2 + \left(\frac{a+2}{a-1}\right)x + b, a \neq 1$. If $f(x + y) = f(x) + f(y) + 1 - \frac{2}{7}xy$, then the value of $28\sum_{i=1}^{5}|f(i)|$ is

(A) 735 (B) 675 (C) 715 (D) 545

Q63. JEE Main 2025 (Online) 24th January Evening Shift

The function $f : (-\infty, \infty) \to (-\infty, 1)$, defined by $f(x) = \frac{2^x - 2^{-x}}{2^x + 2^{-x}}$ is :

(A) One-one but not onto (B) Onto but not one-one

(C) Both one-one and onto (D) Neither one-one nor onto

Q64. JEE Main 2025 (Online) 24th January Morning Shift

Let $f(x) = \frac{2^{x+2}+16}{2^{2x+1}+2^{x+4}+32}$. Then the value of $8\left(f\left(\frac{1}{15}\right) + f\left(\frac{2}{15}\right) + \ldots + f\left(\frac{59}{15}\right)\right)$ is equal to

(A) 108 (B) 92 (C) 118 (D) 102

Q65. JEE Main 2025 (Online) 23rd January Morning Shift

Let $f(x) = \log_e x$ and $g(x) = \frac{x^4 - 2x^3 + 3x^2 - 2x + 2}{2x^2 - 2x + 1}$. Then the domain of $f \circ g$ is

(A) $(0, \infty)$ (B) $[1, \infty)$ (C) $\mathbb{R}$ (D) $[0, \infty)$

Q66. JEE Main 2025 (Online) 22nd January Evening Shift

Let $\mathrm{A} = \{1, 2, 3, 4\}$ and $\mathrm{B} = \{1, 4, 9, 16\}$. Then the number of many-one functions $f : \mathrm{A} \to \mathrm{B}$ such that $1 \in f(\mathrm{A})$ is equal to :

(A) 151 (B) 139 (C) 163 (D) 127

Q67. JEE Main 2024 (Online) 9th April Evening Shift

Let the range of the function $f(x) = \frac{1}{2 + \sin 3x + \cos 3x}, x \in \mathbb{R}$ be $[a, b]$. If α and β are respectively the A.M. and the G.M. of a and b, then $\frac{\alpha}{\beta}$ is equal to

(A) π (B) $\sqrt{\pi}$ (C) $\sqrt{2}$ (D) 2

Q68. JEE Main 2024 (Online) 9th April Morning Shift

If the domain of the function $f(x) = \sin^{-1}\left(\frac{x-1}{2x+3}\right)$ is $\mathbf{R} - (\alpha, \beta)$, then $12\alpha\beta$ is equal to :

(A) 40 (B) 36 (C) 24 (D) 32

Q69. JEE Main 2024 (Online) 8th April Evening Shift

Let $f(x)=\begin{cases}-\mathrm{a} & \text{if} \quad -\mathrm{a} \leq x \leq 0 \\ x+\mathrm{a} & \text{if} \quad 0<x \leq \mathrm{a}\end{cases}$ where $\mathrm{a}>0$ and $\mathrm{g}(x)=(f(|x|)-|f(x)|)/2$. Then the function $g:[-a, a] \rightarrow[-a, a]$ is

(A) neither one-one nor onto.

(B) both one-one and onto.

(C) one-one.

(D) onto

Q70. JEE Main 2024 (Online) 8th April Morning Shift

Let $[t]$ be the greatest integer less than or equal to t. Let A be the set of all prime factors of 2310 and $f: A \rightarrow \mathbb{Z}$ be the function $f(x)=\left[\log_2\left(x^2+\left[\frac{x^3}{5}\right]\right)\right]$. The number of one-to-one functions from A to the range of f is

(A) 20 (B) 120 (C) 25 (D) 24

Q71. JEE Main 2024 (Online) 6th April Evening Shift

Let $f(x)=\frac{1}{7-\sin 5x}$ be a function defined on $\mathbf{R}$. Then the range of the function $f(x)$ is equal to :

(A) $\left[\frac{1}{8}, \frac{1}{5}\right]$

(B) $\left[\frac{1}{7}, \frac{1}{6}\right]$

(C) $\left[\frac{1}{7}, \frac{1}{5}\right]$

(D) $\left[\frac{1}{8}, \frac{1}{6}\right]$

Q72. JEE Main 2024 (Online) 6th April Morning Shift

The function $f(x)=\frac{x^2+2x-15}{x^2-4x+9}, x \in \mathbb{R}$ is

(A) both one-one and onto.

(B) onto but not one-one.

(C) neither one-one nor onto.

(D) one-one but not onto.

Q73. JEE Main 2024 (Online) 5th April Evening Shift

Let $f, g: \mathbf{R} \rightarrow \mathbf{R}$ be defined as :

$f(x)=|x-1|$ and $g(x)=\begin{cases}\mathrm{e}^x, & x \geq 0 \\ x+1, & x \leq 0.\end{cases}$

Then the function $f(g(x))$ is

(A) neither one-one nor onto.

(B) one-one but not onto.

(C) both one-one and onto.

(D) onto but not one-one.

Q74. JEE Main 2024 (Online) 5th April Morning Shift

Let $A=\{1,3,7,9,11\}$ and $B=\{2,4,5,7,8,10,12\}$. Then the total number of one-one maps $f: A \rightarrow B$, such that $f(1)+f(3)=14$, is :

(A) 120 (B) 180 (C) 240 (D) 480

Q75. JEE Main 2024 (Online) 1st February Evening Shift

If the domain of the function

$f(x)=\frac{\sqrt{x^2-25}}{(4-x^2)}+\log_{10}\left(x^2+2x-15\right)$ is $(-\infty,\alpha)\cup[\beta,\infty)$, then $\alpha^2+\beta^3$ is equal to :

(A) 140 (B) 175 (C) 125 (D) 150

Q76. JEE Main 2024 (Online) 1st February Morning Shift

Let $f:\mathbf{R}\to\mathbf{R}$ and $g:\mathbf{R}\to\mathbf{R}$ be defined as

$f(x)=\begin{cases}\log_e x, & x>0\\ e^{-x}, & x\le 0\end{cases}$ and

$g(x)=\begin{cases}x, & x\geqslant 0\\ e^{x}, & x<0\end{cases}$. Then, gof : $\mathbf{R}\to\mathbf{R}$ is :

(A) one-one but not onto (B) neither one-one nor onto

(C) onto but not one-one (D) both one-one and onto

Q77. JEE Main 2024 (Online) 31st January Morning Shift

If $f(x)=\frac{4x+3}{6x-4}, x\neq\frac{2}{3}$ and $(f\circ f)(x)=g(x)$, where $g:\mathbb{R}-\left\{\frac{2}{3}\right\}\to\mathbb{R}-\left\{\frac{2}{3}\right\}$, then $(g\circ g\circ g)(4)$ is equal to

(A) -4 (B) $\frac{19}{20}$ (C) $-\frac{19}{20}$ (D) 4

Q78. JEE Main 2024 (Online) 30th January Evening Shift

If the domain of the function $f(x)=\log_e\left(\frac{2x+3}{4x^2+x-3}\right)+\cos^{-1}\left(\frac{2x-1}{x+2}\right)$ is $(\alpha,\beta]$, then the value of $5\beta-4\alpha$ is equal to

(A) 9 (B) 12 (C) 11 (D) 10

Q79. JEE Main 2024 (Online) 30th January Morning Shift

If the domain of the function $f(x)=\cos^{-1}\left(\frac{2-|x|}{4}\right)+\{\log_e(3-x)\}^{-1}$ is $[-\alpha,\beta)-\{\gamma\}$, then $\alpha+\beta+\gamma$ is equal to :

(A) 11 (B) 12 (C) 9 (D) 8

Q80. JEE Main 2024 (Online) 29th January Morning Shift

If $f(x)=\begin{cases}2+2x, & -1\le x<0\\ 1-\frac{x}{3}, & 0\le x\le 3\end{cases}; g(x)=\begin{cases}-x, & -3\le x\le 0\\ x, & 0<x\le 1\end{cases}$, then range of $(fog)(x)$ is

(A) $[0,1)$ (B) $[0,3)$ (C) $(0,1]$ (D) $[0,1]$

Q81. JEE Main 2024 (Online) 27th January Evening Shift

Let $f : \mathbf{R} - \left\{\frac{-1}{2}\right\} \to \mathbf{R}$ and $g : \mathbf{R} - \left\{\frac{-5}{2}\right\} \to \mathbf{R}$ be defined as $f(x) = \frac{2x+3}{2x+1}$ and $g(x) = \frac{|x|+1}{2x+5}$. Then, the domain of the function fog is :

(A) $\mathbf{R} - \left\{-\frac{7}{4}\right\}$

(B) $\mathbf{R}$

(C) $\mathbf{R} - \left\{-\frac{5}{2}, -\frac{7}{4}\right\}$

(D) $\mathbf{R} - \left\{-\frac{5}{2}\right\}$

Q82. JEE Main 2024 (Online) 27th January Morning Shift

The function $f : \mathbf{N} - \{1\} \to \mathbf{N}$; defined by $f(\mathrm{n}) =$ the highest prime factor of n, is :

(A) one-one only

(B) neither one-one nor onto

(C) onto only

(D) both one-one and onto

Q83. JEE Main 2023 (Online) 13th April Evening Shift

The range of $f(x) = 4\sin^{-1}\left(\frac{x^2}{x^2+1}\right)$ is

(A) $[0, 2\pi]$

(B) $[0, 2\pi)$

(C) $[0, \pi)$

(D) $[0, \pi]$

Q84. JEE Main 2023 (Online) 13th April Morning Shift

For $x \in \mathbb{R}$, two real valued functions $f(x)$ and $g(x)$ are such that, $g(x) = \sqrt{x} + 1$ and $f \circ g(x) = x + 3 - \sqrt{x}$. Then $f(0)$ is equal to

(A) 5

(B) 0

(C) -3

(D) 1

Q85. JEE Main 2023 (Online) 12th April Morning Shift

Let D be the domain of the function $f(x) = \sin^{-1}\left(\log_{3x}\left(\frac{6+2\log_3 x}{-5x}\right)\right)$. If the range of the function $\mathrm{g} : \mathrm{D} \to \mathbb{R}$ defined by $\mathrm{g}(x) = x - [x]$, ($[x]$ is the greatest integer function), is (α, β), then $\alpha^2 + \frac{5}{\beta}$ is equal to

(A) 45

(B) 136

(C) 46

(D) nearly 135

Q86. JEE Main 2023 (Online) 11th April Evening Shift

The domain of the function $f(x) = \frac{1}{\sqrt{[x]^2 - 3[x] - 10}}$ is : (where $[\mathrm{x}]$ denotes the greatest integer less than or equal to x)

(A) $(-\infty, -2) \cup [6, \infty)$

(B) $(-\infty, -3] \cup [6, \infty)$

(C) $(-\infty, -2) \cup (5, \infty)$

(D) $(-\infty, -3] \cup (5, \infty)$

Q87. JEE Main 2023 (Online) 10th April Morning Shift

If $f(x) = \frac{(\tan 1^\circ)x + \log_e(123)}{x\log_e(1234) - (\tan 1^\circ)}, x > 0$, then the least value of $f(f(x)) + f\left(f\left(\frac{4}{x}\right)\right)$ is :

(A) 2

(B) 4

(C) 0

(D) 8

Q88. JEE Main 2023 (Online) 6th April Evening Shift

Let the sets A and B denote the domain and range respectively of the function $f(x)=\frac{1}{\sqrt{\lceil x\rceil-x}}$, where $\lceil x\rceil$ denotes the smallest integer greater than or equal to x. Then among the statements

(S1) : $A\cap B=(1,\infty)-\mathbb{N}$ and

(S2) : $A\cup B=(1,\infty)$

(A) only $(S2)$ is true

(B) only (S1) is true

(C) neither (S1) nor (S2) is true

(D) both (S1) and (S2) are true

Q89. JEE Main 2023 (Online) 1st February Evening Shift

Let $f:\mathbb{R}-\{0,1\}\to\mathbb{R}$ be a function such that $f(x)+f\left(\frac{1}{1-x}\right)=1+x$. Then $f(2)$ is equal to

(A) $\frac{9}{4}$ (B) $\frac{7}{4}$ (C) $\frac{7}{3}$ (D) $\frac{9}{2}$

Q90. JEE Main 2023 (Online) 31st January Evening Shift

Let $f:\mathbb{R}-\{2,6\}\to\mathbb{R}$ be real valued function

defined as $f(x)=\frac{x^2+2x+1}{x^2-8x+12}$.

Then range of f is

(A) $\left(-\infty,-\frac{21}{4}\right]\cup[1,\infty)$

(B) $\left(-\infty,-\frac{21}{4}\right)\cup(0,\infty)$

(C) $\left(-\infty,-\frac{21}{4}\right]\cup[0,\infty)$

(D) $\left(-\infty,-\frac{21}{4}\right]\cup\left[\frac{21}{4},\infty\right)$

Q91. JEE Main 2023 (Online) 31st January Evening Shift

The absolute minimum value, of the function

$f(x)=\left|x^2-x+1\right|+\left[x^2-x+1\right]$,

where $[t]$ denotes the greatest integer function, in the interval $[-1,2]$, is :

(A) $\frac{3}{4}$ (B) $\frac{3}{2}$ (C) $\frac{1}{4}$ (D) $\frac{5}{4}$

Q92. JEE Main 2023 (Online) 31st January Morning Shift

If the domain of the function $f(x)=\frac{[x]}{1+x^2}$, where $[x]$ is greatest integer $\le x$, is $[2,6)$, then its range is

(A) $\left(\frac{5}{37},\frac{2}{5}\right]-\left\{\frac{9}{29},\frac{27}{109},\frac{18}{89},\frac{9}{53}\right\}$

(B) $\left(\frac{5}{37},\frac{2}{5}\right]$

(C) $\left(\frac{5}{26},\frac{2}{5}\right]$

(D) $\left(\frac{5}{26},\frac{2}{5}\right]-\left\{\frac{9}{29},\frac{27}{109},\frac{18}{89},\frac{9}{53}\right\}$

Q93. JEE Main 2023 (Online) 30th January Evening Shift

The range of the function $f(x)=\sqrt{3-x}+\sqrt{2+x}$ is :

(A) $[2\sqrt{2},\sqrt{11}]$ (B) $[\sqrt{5},\sqrt{13}]$ (C) $[\sqrt{2},\sqrt{7}]$ (D) $[\sqrt{5},\sqrt{10}]$

Q94. JEE Main 2023 (Online) 30th January Morning Shift

Suppose $f:\mathbb{R}\to(0,\infty)$ be a differentiable function such that $5f(x+y)=f(x)\cdot f(y),\forall x,y\in\mathbb{R}$. If $f(3)=320$, then $\sum\limits_{n=0}^{5}f(n)$ is equal to :

(A) 6875 (B) 6525 (C) 6575 (D) 6825

Q95. JEE Main 2023 (Online) 29th January Evening Shift

Consider a function $f:\mathbb{N}\to\mathbb{R}$, satisfying $f(1)+2f(2)+3f(3)+\ldots.+xf(x)=x(x+1)f(x);x\geq 2$ with $f(1)=1$. Then $\frac{1}{f(2022)}+\frac{1}{f(2028)}$ is equal to

(A) 8000 (B) 8400 (C) 8100 (D) 8200

Q96. JEE Main 2023 (Online) 29th January Morning Shift

The domain of $f(x)=\frac{\log_{(x+1)}(x-2)}{e^{2\log_e x}-(2x+3)},x\in\mathbb{R}$ is

(A) $(-1,\infty)-\{3\}$ (B) $\mathbb{R}-\{-1,3\}$ (C) $(2,\infty)-\{3\}$ (D) $\mathbb{R}-\{3\}$

Q97. JEE Main 2023 (Online) 29th January Morning Shift

Let $f:R\to R$ be a function such that $f(x)=\frac{x^2+2x+1}{x^2+1}$. Then

(A) $f(x)$ is many-one in $(-\infty,-1)$

(B) $f(x)$ is one-one in $(-\infty,\infty)$

(C) $f(x)$ is one-one in $[1,\infty)$ but not in $(-\infty,\infty)$

(D) $f(x)$ is many-one in $(1,\infty)$

Q98. JEE Main 2023 (Online) 25th January Evening Shift

The number of functions

$f:\{1,2,3,4\}\to\{a\in Z:|a|\leq 8\}$

satisfying $f(n)+\frac{1}{n}f(n+1)=1,\forall n\in\{1,2,3\}$ is

(A) 2 (B) 3 (C) 1 (D) 4

Q99. JEE Main 2023 (Online) 25th January Evening Shift

Let $f:\mathbb{R}\to\mathbb{R}$ be a function defined by $f(x)=\log_{\sqrt{m}}\{\sqrt{2}(\sin x-\cos x)+m-2\}$, for some m, such that the range of f is [0, 2]. Then the value of m is _____

(A) 4 (B) 3 (C) 5 (D) 2

Q100. JEE Main 2023 (Online) 25th January Evening Shift

Let $f(x)=2x^n+\lambda, \lambda \in R, n \in N$, and $f(4)=133, f(5)=255$. Then the sum of all the positive integer divisors of $(f(3)-f(2))$ is

(A) 60 (B) 58 (C) 61 (D) 59

Q101. JEE Main 2023 (Online) 24th January Evening Shift

Let $f(x)$ be a function such that $f(x+y)=f(x).f(y)$ for all $x, y \in \mathbb{N}$. If $f(1)=3$ and $\sum_{k=1}^{n} f(k)=3279$, then the value of n is

(A) 9 (B) 7 (C) 6 (D) 8

Q102. JEE Main 2023 (Online) 24th January Evening Shift

If $f(x)=\frac{2^{2x}}{2^{2x}+2}, x \in \mathbb{R}$, then $f\left(\frac{1}{2023}\right)+f\left(\frac{2}{2023}\right)+\ldots+f\left(\frac{2022}{2023}\right)$ is equal to

(A) 2011 (B) 2010 (C) 1010 (D) 1011

Q103. JEE Main 2022 (Online) 28th July Evening Shift

Let $f(x)=ax^2+bx+c$ be such that $f(1)=3, f(-2)=\lambda$ and $f(3)=4$. If $f(0)+f(1)+f(-2)+f(3)=14$, then λ is equal to :

(A) -4 (B) $\frac{13}{2}$ (C) $\frac{23}{2}$ (D) 4

Q104. JEE Main 2022 (Online) 28th July Morning Shift

Let α, β and γ be three positive real numbers. Let $f(x)=\alpha x^5+\beta x^3+\gamma x, x \in \mathbf{R}$ and $g: \mathbf{R} \rightarrow \mathbf{R}$ be such that $g(f(x))=x$ for all $x \in \mathbf{R}$. If $\mathrm{a}_1, \mathrm{a}_2, \mathrm{a}_3, \ldots, \mathrm{a}_\mathrm{n}$ be in arithmetic progression with mean zero, then the value of $f\left(g\left(\frac{1}{\mathrm{n}} \sum_{i=1}^{\mathrm{n}} f\left(\mathrm{a}_i\right)\right)\right)$ is equal to :

(A) 0 (B) 3 (C) 9 (D) 27

Q105. JEE Main 2022 (Online) 27th July Morning Shift

Let $f, g: \mathbb{N}-\{1\} \rightarrow \mathbb{N}$ be functions defined by $f(a)=\alpha$, where α is the maximum of the powers of those primes p such that p^α divides a, and $g(a)=a+1$, for all $a \in \mathbb{N}-\{1\}$. Then, the function $f+g$ is

(A) one-one but not onto (B) onto but not one-one

(C) both one-one and onto (D) neither one-one nor onto

Q106. JEE Main 2022 (Online) 25th July Evening Shift

The number of bijective functions $f:\{1,3,5,7, \ldots, 99\} \rightarrow\{2,4,6,8, \ldots .100\}$, such that $f(3) \geq f(9) \geq f(15) \geq f(21) \geq \ldots . . f(99)$, is _____.

(A) ${}^{50}P_{17}$ (B) ${}^{50}P_{33}$ (C) $33! \times 17!$ (D) $\frac{50!}{2}$

Q107. JEE Main 2022 (Online) 25th July Morning Shift

The total number of functions,

$f : \{1, 2, 3, 4\} \to \{1, 2, 3, 4, 5, 6\}$ such that $f(1) + f(2) = f(3)$, is equal to :

(A) 60 (B) 90 (C) 108 (D) 126

Q108. JEE Main 2022 (Online) 28th June Morning Shift

Let a function f : N $\to$ N be defined by

$$f(n) = \begin{cases} 2n, & n = 2, 4, 6, 8, \ldots\ldots \\ n - 1, & n = 3, 7, 11, 15, \ldots\ldots \\ \frac{n+1}{2}, & n = 1, 5, 9, 13, \ldots\ldots \end{cases}$$

then, f is

(A) one-one but not onto (B) onto but not one-one

(C) neither one-one nor onto (D) one-one and onto

Q109. JEE Main 2022 (Online) 26th June Evening Shift

Let f : R $\to$ R be defined as f (x) = x $-$ 1 and g : R $-$ {1, $-$1} $\to$ R be defined as $g(x) = \frac{x^2}{x^2-1}$.

Then the function fog is :

(A) one-one but not onto (B) onto but not one-one

(C) both one-one and onto (D) neither one-one nor onto

Q110. JEE Main 2022 (Online) 26th June Morning Shift

Let $f(x) = \frac{x-1}{x+1}$, $x \in R - \{0, -1, 1\}$. If $f^{n+1}(x) = f(f^n(x))$ for all n $\in$ N, then $f^6(6) + f^7(7)$ is equal to :

(A) $\frac{7}{6}$ (B) $-\frac{3}{2}$ (C) $\frac{7}{12}$ (D) $-\frac{11}{12}$

Q111. JEE Main 2022 (Online) 25th June Morning Shift

Let f : N $\to$ R be a function such that $f(x + y) = 2f(x)f(y)$ for natural numbers x and y. If f(1) = 2, then the value of α for which

$\sum_{k=1}^{10} f(\alpha + k) = \frac{512}{3}(2^{20} - 1)$

holds, is :

(A) 2 (B) 3 (C) 4 (D) 6

Q112. JEE Main 2022 (Online) 25th June Morning Shift

Let $f : R \to R$ and $g : R \to R$ be two functions defined by $f(x) = \log_e(x^2 + 1) - e^{-x} + 1$ and $g(x) = \frac{1-2e^{2x}}{e^x}$. Then, for which of the following range of α, the inequality $f\left(g\left(\frac{(\alpha-1)^2}{3}\right)\right) > f\left(g\left(\alpha - \frac{5}{3}\right)\right)$ holds ?

(A) (2, 3) (B) $(-2, -1)$ (C) (1, 2) (D) $(-1, 1)$

Q113. JEE Main 2021 (Online) 1st September Evening Shift

The range of the function,

$f(x) = \log_{\sqrt{5}}\left(3 + \cos\left(\frac{3\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} - x\right) - \cos\left(\frac{3\pi}{4} - x\right)\right)$ is :

(A) $\left(0, \sqrt{5}\right)$ (B) $[-2, 2]$

(C) $\left[\frac{1}{\sqrt{5}}, \sqrt{5}\right]$ (D) [0, 2]

Q114. JEE Main 2021 (Online) 31st August Evening Shift

Let f : N $\to$ N be a function such that f(m + n) = f(m) + f(n) for every m, n$\in$N. If f(6) = 18, then f(2) . f(3) is equal to :

(A) 6 (B) 54 (C) 18 (D) 36

Q115. JEE Main 2021 (Online) 27th July Evening Shift

Let f : R $\to$ R be defined as $f(x + y) + f(x - y) = 2f(x)f(y), f\left(\frac{1}{2}\right) = -1$. Then, the value of $\sum_{k=1}^{20} \frac{1}{\sin(k)\sin(k+f(k))}$ is equal to :

(A) $\text{cosec}^2(21)\cos(20)\cos(2)$ (B) $\sec^2(1)\sec(21)\cos(20)$

(C) $\text{cosec}^2(1)\text{cosec}(21)\sin(20)$ (D) $\sec^2(21)\sin(20)\sin(2)$

Q116. JEE Main 2021 (Online) 25th July Evening Shift

Consider function f : A $\to$ B and g : B $\to$ C (A, B, C $\subseteq$ R) such that $(gof)^{-1}$ exists, then :

(A) f and g both are one-one (B) f and g both are onto

(C) f is one-one and g is onto (D) f is onto and g is one-one

Q117. JEE Main 2021 (Online) 25th July Morning Shift

Let g : N $\rightarrow$ N be defined as

g(3n + 1) = 3n + 2,

g(3n + 2) = 3n + 3,

g(3n + 3) = 3n + 1, for all n $\geq$ 0.

Then which of the following statements is true?

(A) There exists an onto function f : N $\rightarrow$ N such that fog = f

(B) There exists a one-one function f : N $\rightarrow$ N such that fog = f

(C) gogog = g

(D) There exists a function f : N $\rightarrow$ N such that gof = f

Q118. JEE Main 2021 (Online) 20th July Evening Shift

Let $f : R - \left\{\frac{\alpha}{6}\right\} \rightarrow R$ be defined by $f(x) = \frac{5x+3}{6x-\alpha}$. Then the value of α for which (fof)(x) = x, for all $x \in R - \left\{\frac{\alpha}{6}\right\}$, is :

(A) No such α exists

(B) 5

(C) 8

(D) 6

Q119. JEE Main 2021 (Online) 20th July Morning Shift

Let [x] denote the greatest integer $\leq$ x, where x $\in$ R. If the domain of the real valued function $f(x) = \sqrt{\frac{[[x]-2}{[[x]-3}}$ is $(-\infty, a) \cup [b, c) \cup [4, \infty)$, a < b < c, then the value of a + b + c is :

(A) 8

(B) 1

(C) -2

(D) -3

Q120. JEE Main 2021 (Online) 18th March Evening Shift

Let f : R $-$ {3} $\rightarrow$ R $-$ {1} be defined by f(x) = $\frac{x-2}{x-3}$.

Let g : R $\rightarrow$ R be given as g(x) = 2x $-$ 3. Then, the sum of all the values of x for which $f^{-1}(x) + g^{-1}(x) = \frac{13}{2}$ is equal to :

(A) 3

(B) 5

(C) 2

(D) 7

Q121. JEE Main 2021 (Online) 18th March Morning Shift

The real valued function
$f(x) = \frac{\cos ec^{-1}x}{\sqrt{x-[x]}}$, where [x] denotes the greatest integer less than or equal to x, is defined for all x belonging to :

(A) all real except integers

(B) all non-integers except the interval $[-1, 1]$

(C) all integers except 0, -1, 1

(D) all real except the interval $[-1, 1]$

Q122. JEE Main 2021 (Online) 18th March Morning Shift

If the functions are defined as $f(x) = \sqrt{x}$ and $g(x) = \sqrt{1-x}$, then what is the common domain of the following functions :

f + g, f $-$ g, f/g, g/f, g $-$ f where $(f \pm g)(x) = f(x) \pm g(x), (f/g)x = \frac{f(x)}{g(x)}$

(A) $0 \le x \le 1$ (B) $0 \le x < 1$ (C) $0 < x < 1$ (D) $0 < x \le 1$

Q123. JEE Main 2021 (Online) 17th March Morning Shift

The inverse of $y = 5^{\log x}$ is :

(A) $x = 5^{\log y}$ (B) $x = y^{\frac{1}{\log 5}}$ (C) $x = 5^{\frac{1}{\log y}}$ (D) $x = y^{\log 5}$

Q124. JEE Main 2021 (Online) 16th March Morning Shift

The range of a$\in$R for which the

function f(x) = (4a $-$ 3)(x + $\log_e$ 5) + 2(a $-$ 7) $\cot\left(\frac{x}{2}\right)$ $\sin^2\left(\frac{x}{2}\right)$, x $\neq$ 2nπ, n$\in$N has critical points, is :

(A) [1, ∞) (B) $(-3, 1)$

(C) $\left[-\frac{4}{3}, 2\right]$ (D) $(-\infty, -1]$

Q125. JEE Main 2021 (Online) 26th February Evening Shift

Let $A = \{1, 2, 3, \ldots, 10\}$ and $f : A \to A$ be defined as

$$f(k) = \begin{cases} k+1 & if\, k\, is\, odd \\ k & if\, k\, is\, even \end{cases}$$

Then the number of possible functions $g : A \to A$ such that $gof = f$ is :

(A) 5^5 (B) 10^5 (C) 5! (D) $^{10}C_5$

Q126. JEE Main 2021 (Online) 25th February Evening Shift

A function f(x) is given by $f(x) = \frac{5^x}{5^x+5}$, then the sum of the series $f\left(\frac{1}{20}\right) + f\left(\frac{2}{20}\right) + f\left(\frac{3}{20}\right) + \ldots\ldots\ldots + f\left(\frac{39}{20}\right)$ is equal to :

(A) $\frac{39}{2}$ (B) $\frac{19}{2}$ (C) $\frac{49}{2}$ (D) $\frac{29}{2}$

Q127. JEE Main 2021 (Online) 25th February Evening Shift

Let x denote the total number of one-one functions from a set A with 3 elements to a set B with 5 elements and y denote the total number of one-one functions form the set A to the set A $\times$ B. Then :

(A) 2y = 273x (B) y = 91x (C) 2y = 91x (D) y = 273x

Q128. JEE Main 2021 (Online) 25th February Morning Shift

Let f, g : N $\rightarrow$ N such that f(n + 1) = f(n) + f(1) $\forall$ n$\in$N and g be any arbitrary function. Which of the following statements is NOT true?

(A) If g is onto, then fog is one-one (B) f is one-one

(C) If f is onto, then f(n) = n $\forall$n$\in$N (D) If fog is one-one, then g is one-one

Q129. JEE Main 2021 (Online) 24th February Morning Shift

Let f : R $\rightarrow$ R be defined as f (x) = 2x – 1 and g : R - {1} $\rightarrow$ R be defined as g(x) = $\frac{x-\frac{1}{2}}{x-1}$. Then the composition function f(g(x)) is :

(A) one-one but not onto (B) onto but not one-one

(C) both one-one and onto (D) neither one-one nor onto

Q130. JEE Main 2020 (Online) 6th September Evening Slot

For a suitably chosen real constant a, let a

function, $f : R - \{-a\} \rightarrow R$ be defined by

$f(x) = \frac{a-x}{a+x}$. Further suppose that for any real number $x \neq -a$ and $f(x) \neq -a$,

(fof)(x) = x. Then $f\left(-\frac{1}{2}\right)$ is equal to :

(A) $\frac{1}{3}$ (B) –3 (C) $-\frac{1}{3}$ (D) 3

Q131. JEE Main 2020 (Online) 6th September Morning Slot

If f(x + y) = f(x)f(y) and $\sum_{x=1}^{\infty} f(x) = 2$, x, y $\in$ N, where N is the set of all natural number, then the value of $\frac{f(4)}{f(2)}$ is :

(A) $\frac{2}{3}$ (B) $\frac{1}{9}$ (C) $\frac{1}{3}$ (D) $\frac{4}{9}$

Q132. JEE Main 2020 (Online) 2nd September Evening Slot

Let f : R $\rightarrow$ R be a function which satisfies
f(x + y) = f(x) + f(y) $\forall$ x, y $\in$ R. If f(1) = 2 and
g(n) = $\sum_{k=1}^{(n-1)} f(k)$, n $\in$ N then the value of n, for which g(n) = 20, is :

(A) 20 (B) 9 (C) 5 (D) 4

Q133. JEE Main 2020 (Online) 8th January Evening Slot

Let f : (1, 3) $\rightarrow$ R be a function defined by
$f(x) = \frac{x[x]}{1+x^2}$, where [x] denotes the greatest integer $\leq$ x. Then the range of f is

(A) $\left(\frac{2}{5}, \frac{1}{2}\right) \cup \left(\frac{3}{5}, \frac{4}{5}\right]$

(B) $\left(\frac{3}{5}, \frac{4}{5}\right)$

(C) $\left(\frac{2}{5}, \frac{4}{5}\right]$

(D) $\left(\frac{2}{5}, \frac{3}{5}\right] \cup \left(\frac{3}{4}, \frac{4}{5}\right)$

Q134. JEE Main 2020 (Online) 8th January Morning Slot

The inverse function of

f(x) = $\frac{8^{2x}-8^{-2x}}{8^{2x}+8^{-2x}}$, x $\in$ (-1, 1), is :

(A) $\frac{1}{4}\log_e\left(\frac{1-x}{1+x}\right)$

(B) $\frac{1}{4}(\log_8 e)\log_e\left(\frac{1-x}{1+x}\right)$

(C) $\frac{1}{4}(\log_8 e)\log_e\left(\frac{1+x}{1-x}\right)$

(D) $\frac{1}{4}\log_e\left(\frac{1+x}{1-x}\right)$

Q135. JEE Main 2020 (Online) 7th January Morning Slot

If g(x) = x^2 + x - 1 and
(gof) (x) = $4x^2$ - 10x + 5, then f$\left(\frac{5}{4}\right)$ is equal to:

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) -$\frac{1}{2}$ (D) -$\frac{3}{2}$

Q136. JEE Main 2019 (Online) 12th April Morning Slot

For x $\in$ (0, 3/2), let f(x) = $\sqrt{x}$, g(x) = tan x and h(x) = $\frac{1-x^2}{1+x^2}$. If ϕ (x) = ((hof)og)(x), then $\phi\left(\frac{\pi}{3}\right)$ is equal to :

(A) $\tan\frac{7\pi}{12}$ (B) $\tan\frac{11\pi}{12}$ (C) $\tan\frac{\pi}{12}$ (D) $\tan\frac{5\pi}{12}$

Q137. JEE Main 2019 (Online) 10th April Morning Slot

Let f(x) = x^2 , x ∈ R. For any A ⊆ R, define g (A) = { x ∈ R : f(x) ∈ A}. If S = [0,4], then which one of the following statements is not true ?

(A) g(f(S)) ≠ S (B) f(g(S)) = S (C) f(g(S)) ≠ f(S) (D) g(f(S)) = g(S)

Q138. JEE Main 2019 (Online) 10th April Morning Slot

Let f(x) = e^x − x and g(x) = x^2 − x, ∀ x ∈ R. Then the set of all x ∈ R, where the function h(x) = (fog) (x) is increasing, is :

(A) [0, ∞) (B) $\left[-1,-\frac{1}{2}\right]\cup\left[\frac{1}{2},\infty\right)$

(C) $\left[-\frac{1}{2},0\right]\cup[1,\infty)$ (D) $\left[0,\frac{1}{2}\right]\cup[1,\infty)$

Q139. JEE Main 2019 (Online) 9th April Evening Slot

The domain of the definition of the function

$f(x)=\frac{1}{4-x^2}+\log_{10}(x^3-x)$ is

(A) (-1, 0) ∪ (1, 2) ∪ (2, ∞) (B) (-2, -1) ∪ (-1,0) ∪ (2, ∞)

(C) (1, 2) ∪ (2, ∞) (D) (-1, 0) ∪ (1,2) ∪ (3, ∞)

Q140. JEE Main 2019 (Online) 9th April Morning Slot

Let $\sum_{k=1}^{10} f(a+k)=16\left(2^{10}-1\right)$ where the function f satisfies

f(x + y) = f(x)f(y) for all natural numbers x, y and f(1) = 2. then the natural number 'a' is

(A) 2 (B) 16 (C) 4 (D) 3

Q141. JEE Main 2019 (Online) 9th April Morning Slot

If the function f : R − {1, −1} → A defined by

f(x) = $\frac{x^2}{1-x^2}$, is surjective, then A is equal to

(A) R − (−1, 0) (B) R − {−1} (C) R − [−1, 0) (D) [0, ∞)

Q142. JEE Main 2019 (Online) 8th April Evening Slot

Let f(x) = a^x (a > 0) be written as

f(x) = f_1 (x) + f_2 (x), where f_1 (x) is an even function and f_2 (x) is an odd function.

Then f_1 (x + y) + f_1 (x − y) equals

(A) 2f_1 (x)f_1 (y) (B) 2f_1 (x + y)f_1 (x − y)

(C) 2f_1 (x)f_2 (y) (D) 2f_1 (x + y)f_2 (x − y)

Q143. JEE Main 2019 (Online) 8th April Morning Slot

If $f(x) = \log_e\left(\frac{1-x}{1+x}\right), |x| < 1$ then $f\left(\frac{2x}{1+x^2}\right)$ is equal to

(A) $2f(x^2)$ (B) $2f(x)$ (C) $(f(x))^2$ (D) $-2f(x)$

Q144. JEE Main 2019 (Online) 11th January Evening Slot

The number of functions f from {1, 2, 3, ..., 20} onto {1, 2, 3, ..., 20} such that f(k) is a multiple of 3, whenever k is a multiple of 4, is :

(A) $6^5 \times (15)!$ (B) $5^6 \times 15$ (C) $(15)! \times 6!$ (D) $5! \times 6!$

Q145. JEE Main 2019 (Online) 11th January Evening Slot

Let a function f : $(0, \infty) \to (0, \infty)$ be defined by f(x) = $\left|1 - \frac{1}{x}\right|$. Then f is :

(A) not injective but it is surjective (B) neither injective nor surjective

(C) injective only (D) both injective as well as surjective

Q146. JEE Main 2019 (Online) 11th January Morning Slot

Let f : R $\to$ R be defined by f(x) = $\frac{x}{1+x^2}, x \in R$. Then the range of f is :

(A) $\left[-\frac{1}{2}, \frac{1}{2}\right]$ (B) $R - \left[-\frac{1}{2}, \frac{1}{2}\right]$

(C) $(-1, 1) - \{0\}$ (D) $R - [-1, 1]$

Q147. JEE Main 2019 (Online) 11th January Morning Slot

Let $f_k(x) = \frac{1}{k}\left(\sin^k x + \cos^k x\right)$ for k = 1, 2, 3, ... Then for all x $\in$ R, the value of $f_4(x) - f_6(x)$ is equal to

(A) $\frac{1}{4}$ (B) $\frac{5}{12}$ (C) $\frac{-1}{12}$ (D) $\frac{1}{12}$

Q148. JEE Main 2019 (Online) 10th January Evening Slot

Let N be the set of natural numbers and two functions f and g be defined as f, g : N $\to$ N such that

$$f(n) = \begin{cases} \frac{n+1}{2}; & if\ n\ is\ odd \\ \frac{n}{2}; & if\ n\ is\ even \end{cases};$$

and $g(n) = n - (-1)^n$.

Then fog is -

(A) neither one-one nor onto (B) onto but not one-one

(C) both one-one and onto (D) one-one but not onto

Q149. JEE Main 2019 (Online) 9th January Evening Slot

Let A = {x ∈ **R** : x is not a positive integer}.

Define a function f : A → **R** as $f(x) = \frac{2x}{x-1}$,

then f is :

(A) not injective

(B) neither injective nor surjective

(C) surjective but not injective

(D) injective but not surjective

Q150. JEE Main 2019 (Online) 9th January Morning Slot

For $x \in R - \{0, 1\}$, Let $f_1(x) = \frac{1}{x}$, $f_2(x) = 1 - x$

and $f_3(x) = \frac{1}{1-x}$ be three given

functions. If a function, J(x) satisfies

$(f_2 \circ J \circ f_1)(x) = f_3(x)$ then J(x) is equal to :

(A) $f_1(x)$

(B) $\frac{1}{x} f_3(x)$

(C) $f_2(x)$

(D) $f_3(x)$

Q151. JEE Main 2018 (Online) 15th April Evening Slot

Let f: A → B be a function defined as $f(x) = \frac{x-1}{x-2}$, Where A = **R** − {2} and B = **R** − {1}. Then f is :

(A) invertible and $f^{-1}(y) = \frac{3y-1}{y-1}$

(B) invertible and $f^{-1}(y) = \frac{2y-1}{y-1}$

(C) invertible and $f^{-1}(y) = \frac{2y+1}{y-1}$

(D) not invertible

Q152. JEE Main 2017 (Online) 9th April Morning Slot

The function f : **N** → **N** defined by $f(x) = x - 5\left[\frac{x}{5}\right]$, Where **N** is the set of natural numbers and [x] denotes the greatest integer less than or equal to x, is :

(A) one-one and onto

(B) one-one but not onto.

(C) onto but not one-one.

(D) neither one-one nor onto.

Q153. JEE Main 2017 (Online) 8th April Morning Slot

Let $f(x) = 2^{10}.x + 1$ and $g(x) = 3^{10}.x - 1$. If (fog)(x) = x, then x is equal to :

(A) $\frac{3^{10}-1}{3^{10}-2^{-10}}$

(B) $\frac{2^{10}-1}{2^{10}-3^{-10}}$

(C) $\frac{1-3^{-10}}{2^{10}-3^{-10}}$

(D) $\frac{1-2^{-10}}{3^{10}-2^{-10}}$

Q154. JEE Main 2017 (Offline)

The function $f : R \to \left[-\frac{1}{2}, \frac{1}{2}\right]$ defined as

$f(x) = \frac{x}{1+x^2}$, is

(A) invertible

(B) injective but not surjective.

(C) surjective but not injective

(D) neither injective nor surjective.

Q155. JEE Main 2017 (Offline)

Let a, b, c $\in R$. If f(x) = ax^2 + bx + c is such that
a + b + c = 3 and f(x + y) = f(x) + f(y) + xy, $\forall x, y \in R$,

then $\sum_{n=1}^{10} f(n)$ is equal to

(A) 165 (B) 190 (C) 255 (D) 330

Q156. JEE Main 2016 (Online) 9th April Morning Slot

For x $\in$ **R**, x $\neq$ 0, Let f$_0$(x) = $\frac{1}{1-x}$ and
f$_{n+1}$ (x) = f$_0$(f$_n$(x)), n = 0, 1, 2, ...

Then the value of f$_{100}$(3) + f$_1\left(\frac{2}{3}\right)$ + f$_2\left(\frac{3}{2}\right)$ is equal to :

(A) $\frac{8}{3}$ (B) $\frac{5}{3}$ (C) $\frac{4}{3}$ (D) $\frac{1}{3}$

Q157. JEE Main 2016 (Offline)

If $f(x) + 2f\left(\frac{1}{x}\right) = 3x, x \neq 0$, and $S = \{x \in \mathbf{R} : f(x) = f(-x)\}$; then S :

(A) is an empty set.

(B) contains exactly one element.

(C) contains exactly two elements.

(D) contains more than two elements.

Q158. AIEEE 2011

The domain of the function f(x) = $\frac{1}{\sqrt{|x|-x}}$ is

(A) $(0, \infty)$

(B) $(-\infty, 0)$

(C) $(-\infty, \infty) - \{0\}$

(D) $(-\infty, \infty)$

Q159. AIEEE 2009

Let $f(x) = (x+1)^2 - 1, x \geq -1$

Statement - 1 : The set $\{x : f(x) = f^{-1}(x)\} = \{0, -1\}$.

Statement - 2 : f is a bijection.

(A) Statement - 1 is true, Statement - 2 is true; Statement - 2 is a correct explanation for Statement - 1

(B) Statement - 1 is true, Statement - 2 is true; Statement - 2 is **not** a correct explanation for Statement - 1

(C) Statement - 1 is true, Statement - 2 is false

(D) Statement - 1 is false, Statement - 2 is true

Q160. AIEEE 2009

For real x, let f(x) = x^3 + 5x + 1, then

(A) f is one-one but not onto R

(B) f is onto R but not one-one

(C) f is one-one and onto R

(D) f is neither one-one nor onto R

Q161. AIEEE 2008

Let $f : N \to Y$ be a function defined as f(x) = 4x + 3 where
Y = { y $\in$ N, y = 4x + 3 for some x $\in$ N }.
Show that f is invertible and its inverse is

(A) $g(y) = \frac{3y+4}{4}$

(B) $g(y) = 4 + \frac{y+3}{4}$

(C) $g(y) = \frac{y+3}{4}$

(D) $g(y) = \frac{y-3}{4}$

Q162. AIEEE 2007

The largest interval lying in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ for which the function

$f(x) = 4^{-x^2} + \cos^{-1}\left(\frac{x}{2} - 1\right) + \log(\cos x)$,

is defined, is

(A) $\left[-\frac{\pi}{4}, \frac{\pi}{2}\right)$

(B) $\left[0, \frac{\pi}{2}\right)$

(C) $[0, \pi]$

(D) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Q163. AIEEE 2005

Let $f:(-1,1)\to B$, be a function defined by
$f(x)=\tan^{-1}\frac{2x}{1-x^2}$,
then f is both one-one and onto when B is the interval

(A) $\left(0,\frac{\pi}{2}\right)$ (B) $\left[0,\frac{\pi}{2}\right)$
(C) $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ (D) $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

Q164. AIEEE 2005

A real valued function f(x) satisfies the functional equation

f(x - y) = f(x)f(y) - f(a - x)f(a + y)

where a is a given constant and f(0) = 1, f(2a - x) is equal to

(A) - f(x) (B) f(x) (C) f(a) + f(a - x) (D) f(- x)

Q165. AIEEE 2004

The range of the function f(x) = ${}^{7-x}P_{x-3}$ is

(A) {1, 2, 3, 4, 5} (B) {1, 2, 3, 4, 5, 6}
(C) {1, 2, 3, 4} (D) {1, 2, 3}

Q166. AIEEE 2004

If $f:R\to S$, defined by
$f(x)=\sin x-\sqrt{3}\cos x+1$,
is onto, then the interval of S is

(A) [-1, 3] (B) [-1, 1] (C) [0, 1] (D) [0, 3]

Q167. AIEEE 2004

The graph of the function y = f(x) is symmetrical about the line x = 2, then

(A) $f(x)=-f(-x)$ (B) $f(2+x)=f(2-x)$
(C) $f(x)=f(-x)$ (D) $f(x+2)=f(x-2)$

Q168. AIEEE 2004

The domain of the function
$f(x)=\frac{\sin^{-1}(x-3)}{\sqrt{9-x^2}}$

(A) [1, 2] (B) [2, 3) (C) [1, 2) (D) [2, 3]

Q169. AIEEE 2003

The function $f(x) = \log\left(x + \sqrt{x^2+1}\right)$, is

(A) neither an even nor an odd function
(B) an even function
(C) an odd function
(D) a periodic function

Q170. AIEEE 2003

Domain of definition of the function f(x) = $\frac{3}{4-x^2}$ + $\log_{10}(x^3 - x)$, is

(A) (-1, 0)∪(1, 2)∪(2, ∞)
(B) (1, 2)
(C) (-1, 0) ∪ (1, 2)
(D) (1, 2)∪(2, ∞)

Q171. AIEEE 2003

If $f : R \to R$ satisfies f(x + y) = f(x) + f(y), for all x, y ∈ R and f(1) = 7, then $\sum_{r=1}^{n} f(r)$ is

(A) $\frac{7n(n+1)}{2}$
(B) $\frac{7n}{2}$
(C) $\frac{7(n+1)}{2}$
(D) $7n + (n+1)$

Q172. AIEEE 2003

A function f from the set of natural numbers to integers defined by $f(n) = \begin{cases} \frac{n-1}{2}, & when\ n\ is\ odd \\ -\frac{n}{2}, & when\ n\ is\ even \end{cases}$ is

(A) neither one -one nor onto
(B) one-one but not onto
(C) onto but not one-one
(D) one-one and onto both

Q173. AIEEE 2002

The period of $\sin^2\theta$ is

(A) π^2
(B) π
(C) 2π
(D) $\pi/2$

Q174. AIEEE 2002

Which one is not periodic?

(A) $|\sin 3x| + \sin^2 x$
(B) $\cos\sqrt{x} + \cos^2 x$
(C) $\cos\ 4x + \tan^2 x$
(D) $cos\ 2x + \sin x$

Q175. AIEEE 2002

The domain of $\sin^{-1}\left[\log_3\left(\frac{x}{3}\right)\right]$ is

(A) [1, 9]
(B) [-1, 9]
(C) [9, 1]
(D) [-9, -1]

Answer Key

Numerical

1. **395**	2. **72**	3. **4**	4. **96**	5. **24**
6. **1010**	7. **96**	8. **18**	9. **1024**	10. **44**
11. **360**	12. **180**	13. **20**	14. **432**	15. **3240**
16. **3125**	17. **10**	18. **2039**	19. **50**	20. **1440**
21. **10620**	22. **25**	23. **18**	24. **3395**	25. **37**
26. **190**	27. **99**	28. **2**	29. **31**	30. **490**
31. **720**	32. **0**	33. **2**	34. **5**	35. **19**

MCQ (Single Correct Answer)

36. **D**	37. **B**	38. **A**	39. **B**	40. **A**
41. **B**	42. **C**	43. **C**	44. **A**	45. **D**
46. **B**	47. **D**	48. **B**	49. **D**	50. **D**
51. **C**	52. **D**	53. **A**	54. **A**	55. **B**
56. **D**	57. **A**	58. **D**	59. **A**	60. **C**
61. **D**	62. **B**	63. **A**	64. **C**	65. **C**
66. **A**	67. **C**	68. **D**	69. **A**	70. **B**
71. **D**	72. **C**	73. **A**	74. **C**	75. **D**
76. **B**	77. **D**	78. **B**	79. **A**	80. **D**
81. **D**	82. **B**	83. **B**	84. **A**	85. **D**
86. **A**	87. **B**	88. **B**	89. **A**	90. **C**
91. **A**	92. **B**	93. **D**	94. **D**	95. **C**
96. **C**	97. **C**	98. **A**	99. **C**	100. **A**
101. **B**	102. **D**	103. **D**	104. **A**	105. **D**
106. **B**	107. **B**	108. **D**	109. **D**	110. **B**
111. **C**	112. **A**	113. **D**	114. **B**	115. **C**
116. **C**	117. **A**	118. **B**	119. **C**	120. **B**
121. **B**	122. **C**	123. **B**	124. **C**	125. **B**
126. **A**	127. **C**	128. **A**	129. **A**	130. **D**
131. **D**	132. **C**	133. **A**	134. **C**	135. **C**
136. **B**	137. **D**	138. **D**	139. **A**	140. **D**
141. **C**	142. **A**	143. **B**	144. **C**	145. **B**

146. **A**	147. **D**	148. **B**	149. **D**	150. **D**
151. **B**	152. **D**	153. **D**	154. **C**	155. **D**
156. **B**	157. **C**	158. **B**	159. **C**	160. **C**
161. **D**	162. **B**	163. **D**	164. **A**	165. **D**
166. **A**	167. **B**	168. **B**	169. **C**	170. **A**
171. **A**	172. **D**	173. **B**	174. **B**	175. **A**